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Paper Id: 233578 Roll No.

B.TECH.

(SEM I) THEORY EXAMINATION 2022-23

EMERGING TECHNOLOGY FOR ENGINEERING

Time: 3 Hours ***Total Marks: 25***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *any SIX* questions in brief. 1 x 6 = 6

(a) Write down the major components of IoT System.

(b) Compare between Internet and Intranet.

(c) Write down the key characteristics of blockchain technology.

(d) Differentiate between additive manufacturing and subtractive manufacturing.

(e) What are the procedures for becoming a drone pilot in India?

(f) Write down the key objectives of 5G.

(g) Compare and contrast between cloud computing and distributed computing.

(h) What is Artificial Intelligence?

SECTION B

2. Attempt any *three* of the following: 3x3=9

(a) Define Sensors. Explain their types and use in detail.

(b) What is Cloud computing? Discuss the various applications of it.

(c) What are Smart contracts? Explain the features and applications of it.

(d) Define Augmented Reality (AR). Compare and contrast between AR and VR.

(e) Discuss the various applications of 3D printing in detail.

SECTION C

3. Attempt any *one* part of the following: 2x1=2

(a) What do you mean by Industrial Internet of Things (IIoT)? Explain with some applications.

(b) Discuss the various layers of IoT in detail.

4. Attempt any *one* part of the following: 2x1=2

(a) Write a short note on AWS and why it is so popular?

(b) Discuss Microsoft Azure with its features and services in detail.

5. Attempt any *one* part of the following: 2x1=2

(a) What is cryptocurrency? Why cryptocurrencies are volatile in nature?

(b) Discuss the various applications of blockchain technology in detail.

6. Attempt any *one* part of the following: 2x1=2

(a) Discuss about the 3D printing process in detail. What are the advantages and limitations of 3D printing?

(b) What is unmanned aerial vehicle? Discuss the different types of it in detail.

7. Attempt any *one* part of the following: 2x1=2

(a) What is BCI? Explain the working and major challenges of BCI in detail.

(b) Compare and contrast 4G and 5G. How 5G will boostvirtual reality?

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